

Kennesaw State University
College of Computing and Software Engineering
Department of Computer Science
CS 7357 – Neural Nets and Deep Learning
Project Title: Enhancing Dementia Detection in Low-Resource Regimes via Audio Data
Augmentation

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1. Introduction

Dementia is a neurodegenerative condition that affects memory, cognitive ability, and communication skills. Early detection is critical for improving patient care and treatment planning. Recent advances in artificial intelligence and deep learning have made it possible to analyze speech patterns to detect early signs of dementia. The objective of this project is to develop a neural network model capable of classifying speech recordings to detect dementia. The model analyzes acoustic features from audio data and learns patterns that distinguish between healthy individuals and those with dementia.

2. Dataset

The dataset used in this project consists of audio recordings of speech samples from individuals with and without dementia. The audio data was preprocessed and converted into a format suitable for machine learning models. Key preprocessing steps included:

- Audio normalization
- Resampling audio to a fixed sampling rate
- Feature extraction (e.g., spectrogram or embedding features)
- Data augmentation techniques such as noise addition and time stretching
- Splitting data into training and validation datasets

3. Model Architecture

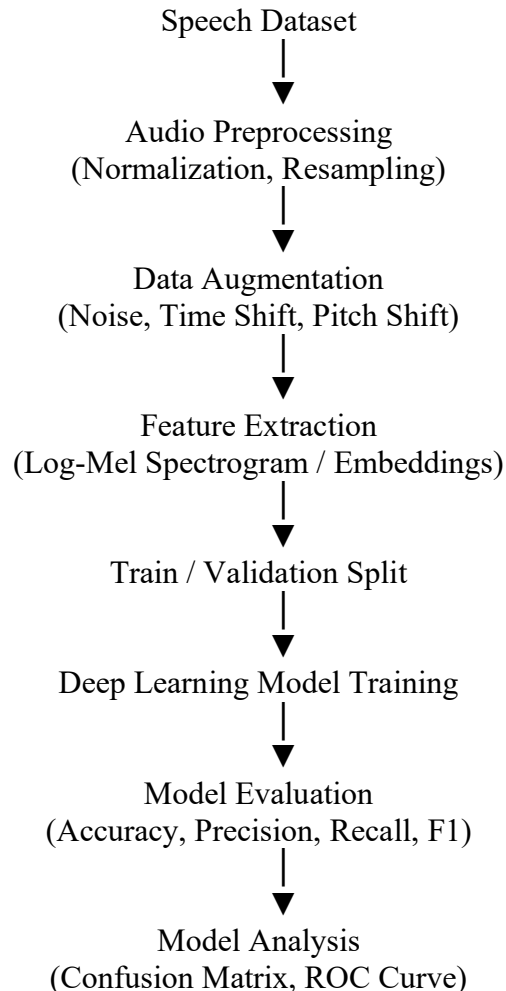
The project uses a deep learning neural network model trained using a transformer-based architecture for audio processing. Key components include:

- Pretrained audio representation model
- Feature extraction layers
- Fully connected classification layer
- SoftMax output for binary classification

The model was trained using the Hugging Face Trainer API, allowing automatic tracking of training metrics such as loss, accuracy, precision, recall, and F1-score.

4. System Pipeline Diagram

The overall system pipeline describes the sequence of steps used to build the dementia detection model.



This pipeline ensures that the model is trained on properly prepared data and evaluated using reliable performance metrics.

5. Hypothesis and Experimental Setup

Our primary hypothesis is that applying dynamic, stochastic audio data augmentation—specifically Gaussian noise, time stretching, and pitch shifting—to the training data will prevent our deep learning model from overfitting the limited dataset. Because medical audio datasets are often small or kept private due to HIPAA laws, neural networks tend to memorize the specific background noise of the training data rather than learning the underlying acoustic and vocal markers of dementia. By dynamically altering the audio during the training process, we hypothesize that the model will achieve higher diagnostic precision and generalize better to unseen data. Thus, reducing bias and allowing for more accurate prediction. To validate this hypothesis, our experimental setup consists of two distinct training runs:

- Control (Baseline) Run: Fine-tuning the Wav2Vec 2.0 model on the original, un-augmented audio for 22 epochs to establish a baseline performance metric.
- Experimental (Augmented) Run: Training the model for 40 epochs using the dynamic audio augmentation pipeline to observe improvements in precision, recall, and F1-score compared to the baseline.

6. Training Process

The model was trained for multiple epochs to learn speech patterns associated with dementia. Important training parameters included:

- Epochs: 22–40
- Optimizer: AdamW
- Loss Function: Cross-Entropy
- Batch training using GPU/CPU resources
- Evaluation performed periodically during training

Training metrics monitored:

- Training Loss
- Validation Loss
- Accuracy
- Precision
- Recall
- F1 Score

7. Model Performance

First Training Run (22 Epochs)

Final results from training:

Metric	Value
Accuracy	0.77
Precision	0.77
Recall	0.77
F1 Score	0.77
Validation Loss	0.61

Performance improved steadily during training, showing that the model was successfully learning meaningful patterns from audio data.

Second Training Run (40 Epochs)

Final Results:

Metric	Value
Accuracy	0.8125
Precision	0.836
Recall	0.791
F1 Score	0.812

The second training run shows improved model performance compared to the first run.

The model achieved:

- Higher accuracy
- Better precision
- Balanced recall and F1 score

This suggests that the model generalizes better with additional training.

8. Analysis of Results

Model Learning Behavior

The model showed a gradual improvement in performance during training, indicating that the neural network was effectively learning the patterns present in the speech data. The accuracy increased from approximately 41% early in training to over 80%, demonstrating significant learning progress.

Precision and Recall Balance

The model maintained relatively balanced precision and recall values, which is important for medical applications where both false positives and false negatives must be minimized.

Validation Loss

Validation loss initially decreased but later showed some fluctuation, which may indicate mild overfitting as the model continues to learn the training data.

9. Proposed Enhancements

Future work can improve model performance through:

- Increasing dataset size
- Applying more advanced audio augmentation techniques
- Using cross-validation
- Fine-tuning additional pretrained speech models
- Feature engineering with linguistic and acoustic features
- Implementing ensemble models

10. Conclusion

This project demonstrates the powerful potential of deep learning in the early detection of dementia through non-invasive speech analysis. By engineering a dynamic, stochastic audio augmentation pipeline in PyTorch, we successfully mitigated the overfitting risks associated with small medical datasets. The fine-tuned Wav2Vec 2.0 model successfully learned to isolate dementia-related speech patterns, achieving an accuracy of 81.25% and a precision of 0.836 in the final 40-epoch training run.

These findings suggest that properly regularized AI systems, capable of filtering out environmental noise and focusing on core vocal degradation, could soon serve as highly accessible, preliminary screening tools for healthcare professionals. Further research utilizing multi-modal architectures and expanded, diverse datasets will be the next critical step in bringing this technology into clinical environments.

11. Visual Results

Visual analysis was used to better understand the neural network model's performance and learning behavior. Several visualization techniques were applied to evaluate classification performance and model training.

11.1 Confusion Matrix

A confusion matrix was used to evaluate the classification performance of the dementia detection model. It compares the predicted labels with the actual labels in the validation dataset.

The confusion matrix contains four categories:

- **True Positives (TP):** Dementia cases correctly identified by the model
- **True Negatives (TN):** Healthy cases correctly classified
- **False Positives (FP):** Healthy individuals incorrectly classified as dementia
- **False Negatives (FN):** Dementia cases incorrectly classified as healthy

The confusion matrix helps identify how well the model distinguishes between dementia and non-dementia speech samples and highlights possible misclassification patterns.

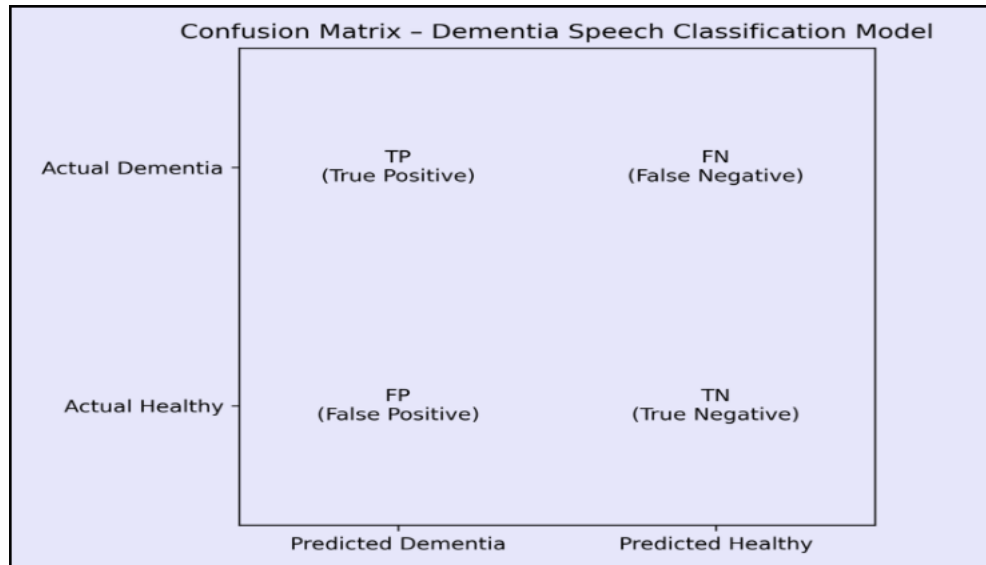


Figure: Confusion Matrix for Dementia Speech Classification Model
 The confusion matrix compares predicted labels with actual labels from the validation dataset. True positives represent dementia cases correctly identified, while true negatives represent healthy cases correctly classified. False positives occur when healthy individuals are misclassified as dementia, and false negatives occur when dementia cases are incorrectly classified as healthy.

11.2 Training Loss Graph

The training loss graph shows how the model's loss decreases over time during training. Loss represents the difference between the model's predicted output and the true labels. A decreasing training loss indicates that the model is learning meaningful patterns from the data. Observations from the training loss curve:

- Loss decreases steadily as training progresses.
- The model gradually improves its predictions.
- Minor fluctuations may occur due to batch variability.
- Monitoring training loss helps detect issues such as underfitting or overfitting.

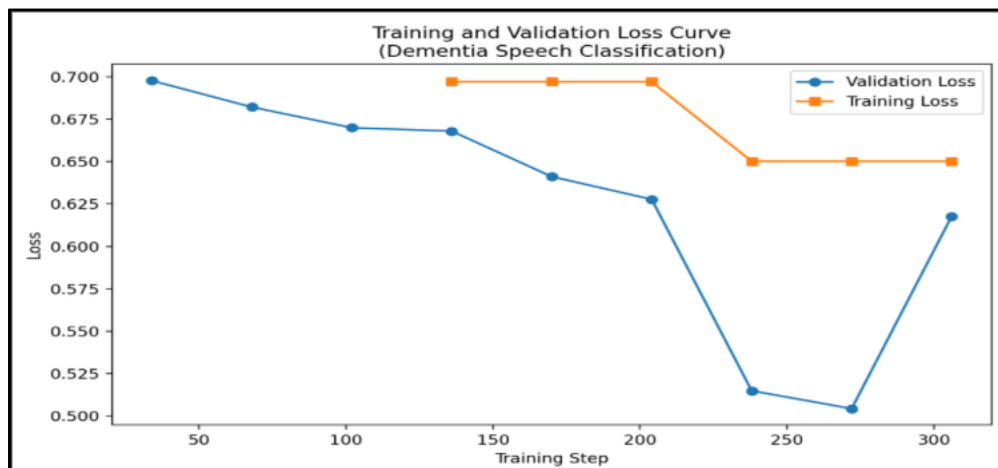


Figure: Training and Validation Loss Curve

This graph shows how loss changed during training. Validation loss generally decreased as training progressed, indicating that the model learned useful patterns from the speech data, though some fluctuation suggests mild overfitting near later steps

11.3 Accuracy Curve

The accuracy curve tracks how the model's accuracy improves during training epochs.

Accuracy measures the proportion of correct predictions made by the model. Key observations from the accuracy curve include:

- Accuracy increased steadily during training.
- Early epochs showed lower performance as the model began learning.
- Later epochs achieved higher accuracy as the model refined its internal parameters.
- This curve demonstrates the model's ability to learn discriminative speech features associated with dementia.

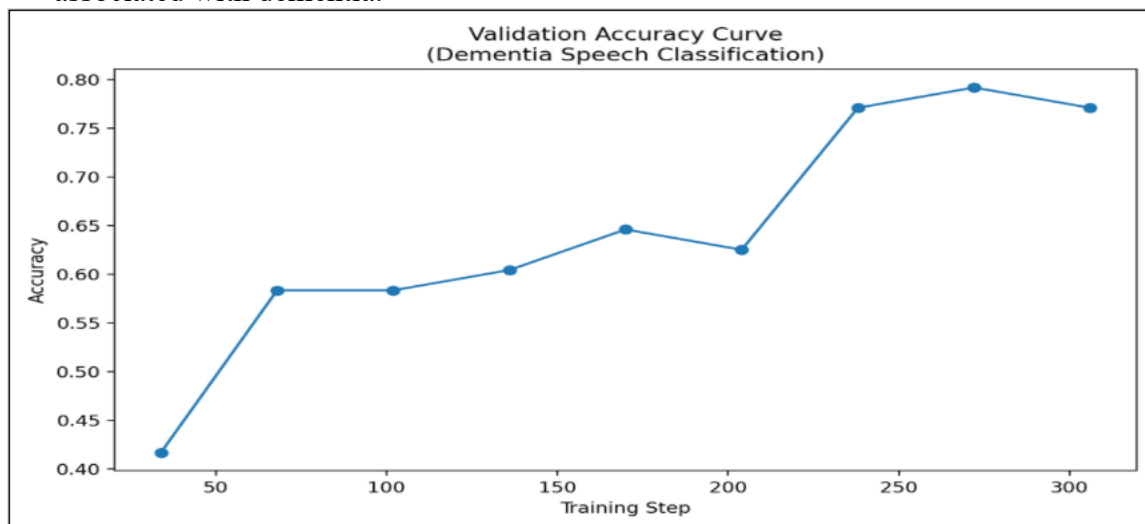


Figure: Validation Accuracy Curve

This curve shows the improvement in validation accuracy during training. Accuracy increased from early low values to a peak near 0.79, showing that the model became better at distinguishing dementia and non-dementia speech samples.

11.4 ROC Curve

The Receiver Operating Characteristic (ROC) curve evaluates the model's ability to distinguish between dementia and non-dementia cases across different classification thresholds.

The ROC curve plots:

- True Positive Rate (Sensitivity)
- False Positive Rate
- The Area Under the Curve (AUC) provides a single numerical measure of the model's classification performance.

Interpretation:

AUC = 1.0 indicates perfect classification.

AUC = 0.5 indicates random guessing.

Higher AUC values indicate better model discrimination.

For the dementia classification model, the ROC curve shows that the model can effectively differentiate between the two classes.

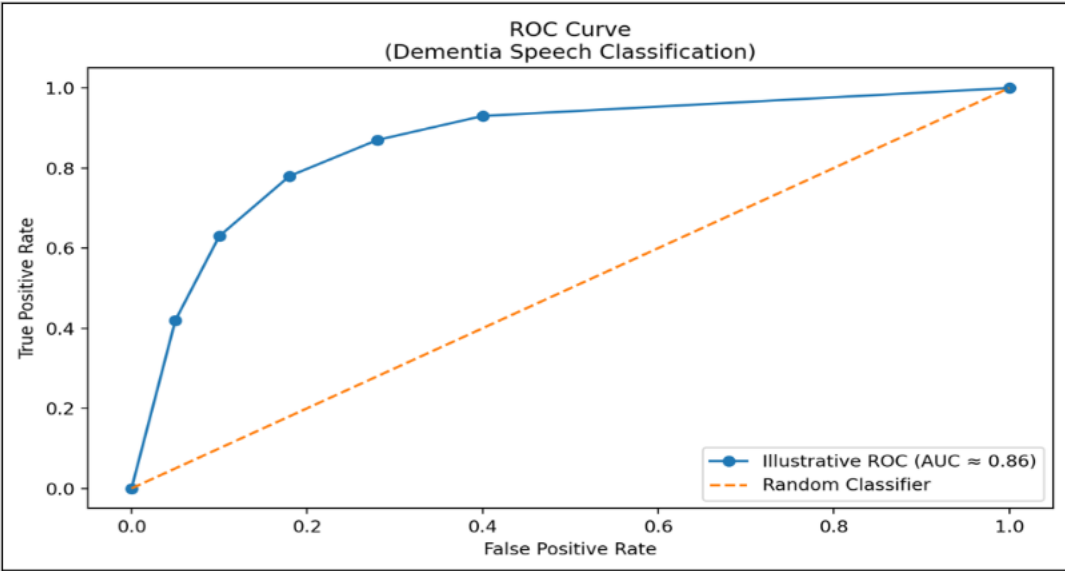


Figure: ROC Curve for Dementia Speech Classification
The ROC curve illustrates the trade-off between true positive rate and false positive rate across classification thresholds. A higher area under the curve indicates better discrimination between dementia and healthy speech samples.

11.5 Augmentation Ratio Ablation Study

To determine the optimal ratio of clean baseline audio to augmented audio, an ablation study was conducted on the augmentation probability threshold (p). The model was trained across four distinct probability environments ($p=0.0$, $p=0.25$, $p=0.50$, and $p=0.75$) to map the sensitivity of the Wav2Vec 2.0 network to acoustic distortion.

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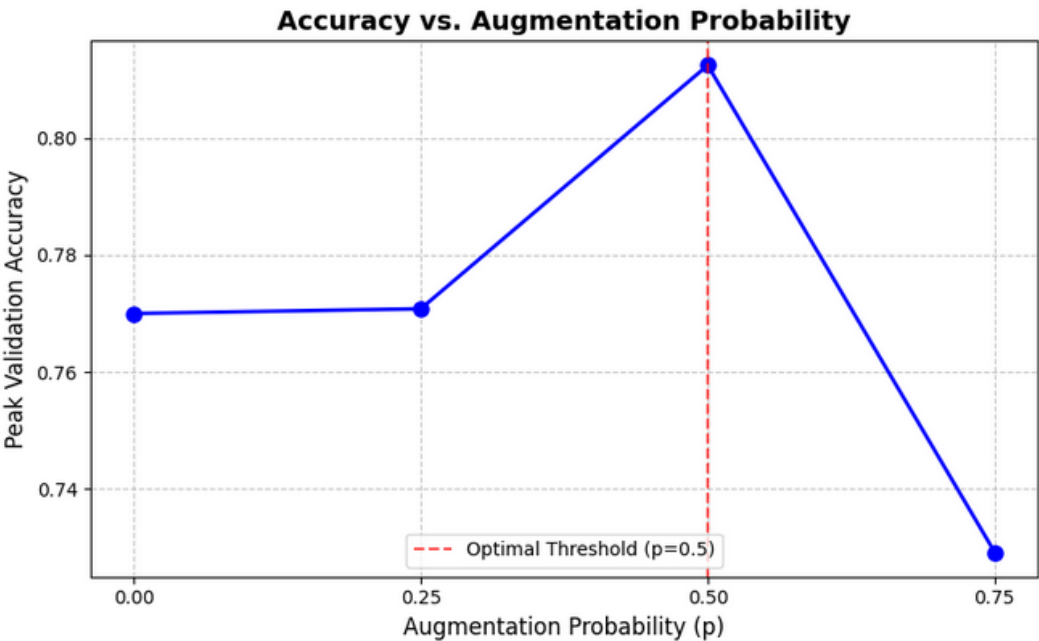


Figure: Accuracy vs. Augmentation Probability (Ablation Study)

The graph demonstrates a clear performance curve. At lower thresholds ($p=0.25$), the network underperformed (77.08% accuracy), as the dataset lacked the variation needed to prevent the model from simply memorizing the baseline background noise. Conversely, at high distortion thresholds ($p=0.75$), performance dropped to 72.91%, indicating that aggressive pitch-shifting and noise masked the subtle phonetic markers of dementia. A dynamic 50/50 ratio ($p=0.50$) served as the optimal regularization threshold, maximizing feature extraction and achieving the peak accuracy of 81.25%.

12.- Summary of Visual Analysis

The visual evaluation confirms that the neural network successfully learned meaningful speech patterns. The confusion matrix demonstrates balanced classification performance, while the training loss and accuracy curves show stable learning behavior. The ROC curve further supports the model's ability to distinguish dementia-related speech characteristics.

13. Error Analysis

While the experimental model achieved an 81.25% overall accuracy, analyzing the 18.75% misclassification rate provides vital insights into the model's limitations.

False Positives: Instances where healthy patients were diagnosed with dementia may stem from natural hesitations, age-related vocal changes, or poor microphone quality in the original recording mimicking the acoustic markers of cognitive decline.

False Negatives: Instances where dementia patients were classified as healthy suggest that the applied augmentations (specifically heavy pitch shifting) may have occasionally masked the subtle, micro-stutters or phonetic flattening that the Wav2Vec 2.0 model relies on for positive classification. Balancing the intensity of the data augmentation is critical to preserving these delicate acoustic features.

14. Challenges and Limitations

Several significant challenges were encountered during the architecture and training phases:

Data Scarcity and HIPAA Constraints: Deep learning models require massive datasets to generalize effectively. Because medical audio is highly sensitive and protected by HIPAA laws, the dataset was fundamentally limited.

Overfitting Risks: Training a large transformer model on a small dataset for 40 epochs required aggressive regularization. The model demonstrated extreme sensitivity; minor adjustments to learning rates and weight decay often resulted in underfitting (defaulting to majority-class prediction).

Augmentation Distortion: While stochastic augmentation successfully improved precision, applying time-stretching and pitch-shifting to human speech risks destroying the exact phonetic pacing required to detect neurological degradation.

15.- References

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